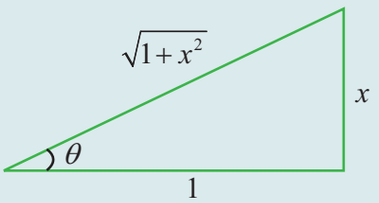
$$= \int \tan^{-1}(\tan 2\theta)\sec^2\theta\, d\theta$$
$$= \int 2\theta \sec^2\theta\, d\theta$$
$$= 2\int(\theta)(\sec^2\theta\, d\theta)$$



Applying integration by parts

$$I = 2\left[\theta\tan\theta - \int\tan\theta d\theta\right] \qquad \tan\theta = x$$
$$= 2(\theta\tan\theta - \log|\sec\theta|) + c \qquad \sec\theta = \sqrt{1+x^2}$$
$$\int \tan^{-1}\left(\frac{2x}{1-x^2}\right)dx = 2x\tan^{-1}x - 2\log|\sqrt{1+x^2}| + c$$

11.7.6 Bernoulli's formula for Integration by Parts

If u and v are functions of x, then the Bernoulli's rule is

$$\int udv = uv - u'v_1 + u''v_2 - \cdots$$

where $u', u'', u''', \ldots$ are successive derivatives of u and

$v, v_1, v_2, v_3, \cdots$ are successive integrals of dv

Bernoulli's formula is advantageously applied when $u = x^n$ (n is a positive integer)

For the following problems we have to apply the integration by parts two or more times to find the solution. In this case Bernoulli's formula helps to find the solution easily.

Example 11.35

Integrate the following with respect to x.

(i) x^2e^{5x} (ii) $x^3\cos x$ (iii) x^3e^{-x}

Solution

(i) $\int x^2e^{5x}dx.$

Applying Bernoulli's formula

$$\int udv = uv - u'v_1 + u''v_2 - \cdots$$

	$dv = e^{5x}dx$
$u = x^2$	$v = \frac{e^{5x}}{5}$
$u' = 2x$	$v_1 = \frac{e^{5x}}{5^2}$
$u'' = 2$	$v_2 = \frac{e^{5x}}{5^3}$
$u''' = 0$	$v_3 = \frac{e^{5x}}{5^4}$

$$\int x^2e^{5x}\,dx = \left(x^2\right)\left(\frac{e^{5x}}{5}\right) - (2x)\left(\frac{e^{5x}}{5^2}\right) + (2)\left(\frac{e^{5x}}{5^3}\right) - (0)\left(\frac{e^{5x}}{5^4}\right) + 0 + \cdots + 0 + c$$

$$= \frac{x^2e^{5x}}{5} - \frac{2xe^{5x}}{25} + \frac{2e^{5x}}{125} + c.$$

(ii) $\int x^3\cos x\,dx.$

Applying Bernoulli's formula

$$\int udv = uv - u'v_1 + u''v_2 - \cdots$$

	$dv = \cos x\,dx$
$u = x^3,$	$v = \sin x$
$u' = 3x,^2$	$v_1 = -\cos x$
$u'' = 6x,$	$v_2 = -\sin x$
$u''' = 6,$	$v_3 = \cos x$

$$\int x^3\cos x\,dx = \left(x^3\right)(\sin x) - \left(3x^2\right)(-\cos x) + (6x)(-\sin x) - (6)(\cos x) + c$$
$$= x^3\sin x + 3x^2\cos x - 6x\sin x - 6\cos x + c.$$

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